SHARVIN TAGUNDE

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SUMMARY

Mechanical Engineering graduate student with 1 year + of hands-on CAD & Manufacturing experience across two engineering internships and projects – producing engineering drawings, layouts and technical documentation for industrial and control applications. Proficient in AutoCAD & SolidWorks with a certified background in drafting schematics & experienced in BOM generation, vendor drawing review and meeting compliance requirements. Eager to support and contribute to protection & control system design within critical power infrastructure.

EDUCATION

University of Texas at Dallas *Dallas, Texas*
*Master of Science (MS). Mechanical Engineering | GPA: **3.75**/4.0* Aug 2024 – May 2026

Savitribai Phule Pune University *Pune, India*
*Bachelor of Engineering (B.E). Mechanical Engineering | GPA: **3.73**/4.0* June 2020 – July 2023

TECHNICAL SKILLS

CAD Design & Drafting: AutoCAD, SolidWorks, Fusion 360, PTC Creo – parts, assembly, technical drawings, schematics, wiring diagrams & layouts.
Engineering: Product Design, GD&T, FEA, DFM, BOMs, Technical documentation, Design of Experiment, 3D Printing, Optimization, & Thermal analysis.
Simulation: ANSYS Workbench, Fluent CFD, Abaqus CAE, MATLAB, Python, Solid CAM.
Certifications: AutoCAD: Drafting Blueprints & Schematics | AI for Mechanical Engineers | Six Sigma Yellow Belt

GRADUATE RESEARCH

Electromagnetic Wave Energy Harvesting - HBS Lab, UT Dallas Aug 2024 – Present

Working on development of a novel low-frequency wave energy harvesting system targeting efficient power generation from irregular & ultra-low-frequency ocean waves. My work aims to create a robust, & scalable solution using iterative design, prototype development, and system-level characterization thus addressing key limitations of conventional energy harvesting systems, with potential applications in offshore electronic sensing and autonomous marine devices.

WORK EXPERIENCE

University of Texas at Dallas *Texas, USA*
Student Assistant | Mechanical Engineering Grader Aug 2025 – Current

- Evaluating undergrad coursework for CAD, Robotics, FEA, & Applied Engineering Math, giving precise technical feedback on course assessments.

Small Spark Concepts *Pune, India*
Product Design Intern | Clean Tech Startup Sep 2023 – May 2024

- Redesigned automotive air filter mold geometries using DFM principles and iterative CAD drafting, improving part manufacturability and reducing complexity for production-scale deployment. Generated bills of materials (BOMs) and part numbering documentation for air filter molds and components, streamlining procurement, inventory classification, and production planning across the product line using ERP software.
- Integrated Lean and Agile methodologies including Value Stream Mapping and Kanban — into manufacturing workflows, identifying and eliminating non-value-added process steps and refining designs while also ensuring quality control, thus reducing overall production lead time by an estimated 10 – 15% and accelerating time-to-ship for the air filter product line.

OM Engineering *Pune, India*
Engineering Design Intern | Specialized Machinery Manufacturer Mar 2020 – Aug 2020

- Produced multi-part CAD models & production-ready technical drawings with full GD&T annotations and job estimates for industrial machine setups, enabling direct handoff to machinists with zero revision cycles on primary designs. Reviewed vendor supplied drawings & integrated component specifications & control diagrams into final design package, ensuring dimensional accuracy & compatibility across machines.
- Conducted interference studies and static structural analyses to validate component fit and integrity; optimized machining sequences and process plans in collaboration with machinists and engineers, improving first-pass assembly to ensure installation readiness.

ACADEMIC PROJECTS

Multi-Objective Optimization — Pin Fin Heat Sink *(Academic Project):* *Mar 2026*

- Built a physics-based analytical model and generated 200-sample DOE datasets; validated designs via ANSYS Fluent CFD with <20% deviation from surrogate predictions across Aluminum and Silicon material sets.
- Integrated surrogate models (ANN, Polynomial) into a custom EA algorithm NSGA-II framework; applied classification criteria to extract optimal Pareto trade-offs of the heat sink design for electronics thermal management applications.

Precision Bush Insertion Machine — Engine Head Assembly *(Industrial Project):* *Sep 2023*

- Designed and fabricated a sensor-integrated bush insertion machine for engine head manufacturing; reviewed vendor-supplied actuator and sensor drawings and integrated them into the final design package to ensure positional accuracy and reliable, repeatable insertion cycles.
- Led end-to-end machine design from concept through fabrication, CAD, assembly documentation, and layouts to guide fabrication and troubleshooting while collaborating with machinists and supervisor to resolve design and fitment challenges, ensuring the machine setup was completed, validated, and ready for installation within the project deadline."